

Algorithm update for GOCI-II remote-sensing reflectance product v2024-1

Korea Hydrographic and Oceanographic Agency(National Ocean Satellite Center)
and Korea Institute of Ocean Science and Technology(Korea Ocean Satellite Center)

1. Background

National Ocean Satellite Center of Korea Hydrographic and Oceanographic Agency and the Korea Institute of Ocean Science and Technology's Ocean Satellite Center are continuously conducting rigorous calibration and validation to enhance the accuracy of the Geostationary Korea Multi Purpose Satellite-2B(GOCI-II). In relation to this, updates on the accuracy improvement of GOCI-II are being carried out.

2. Updated features

In this update, we focused on improving accuracy of a remote-sensing reflectance (R_{rs}) product of GOCI-II. The improvement of R_{rs} is expected to enhance overall quality of the GOCI-II ocean color products including chlorophyll-a concentration, total suspended material concentration, absorption coefficient of dissolved organic matter, and inherent optical properties i.e., absorption and scattering coefficients. The updated features to improve the GOCI-II R_{rs} are correction of the sensor degradation and update of the system vicarious calibration gains.

2.1 Correction of the GOCI-II's sensor degradation

To enhance a temporal consistency of the GOCI-II's radiometric calibration, the sensor degradation correction model is established by monitoring the Solar Diffuser (SD) and the Diffuser Aging Monitoring Device (DAMD) after the BTDF correction as shown in figure 1.

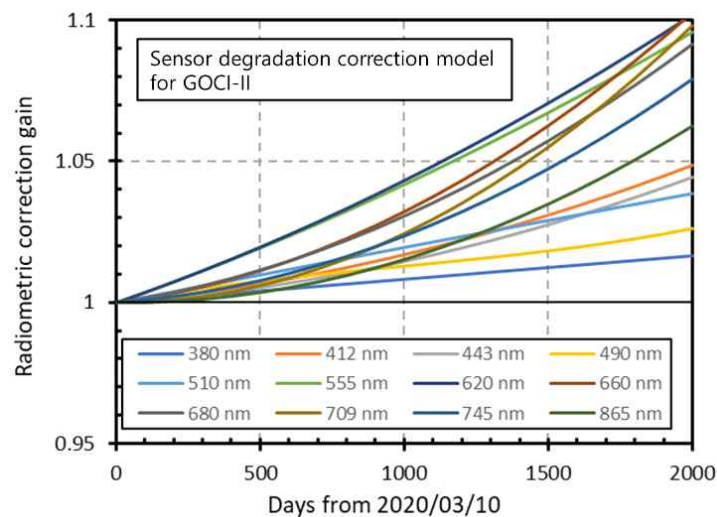


Fig 1. Coefficients for the sensor degradation of GOCI-II

2.2 Adjustment of the system vicarious calibration gains for GOCI-II

A system vicarious calibration has been most widely applied for ocean color satellite missions to enhance agreement between atmospheric correction with radiometric calibration system and actual observations (Franz et al., 2007). The vicarious calibration for the GOCI-II atmospheric correction system is similar with the traditional approach (Franz et al., 2007; Ahn et al., 2015), however, the two-NIR bands are calibrated by directly comparing with the previous GOCI's TOA radiance. Moreover, the GOCI-II's 380~709 nm bands are calibrated using VIIRS's R_{rs} processed by OBP/NASA in the open ocean with spectral conversion by case-1 ocean color model. The updated vicarious gains for GOCI-II are summarized in table 1 and figure 2.

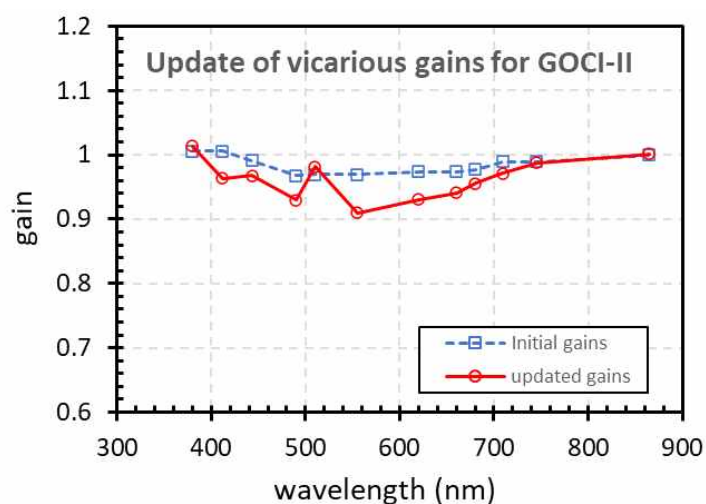


Fig 2. Comparison of the initial phase vicarious gains (blue rectangle) and the updated vicarious gains (red circle).

Table 1. Tabulated vicarious gains for GOCI-II

Band (nm)	380	412	443	490	510	555	620	660	680	709	745	865
Gain	1.013	0.960	0.968	0.930	0.981	0.910	0.930	0.941	0.956	0.971	0.988	1.001

2.3 Improvement for highly turbid waters

The GOCI-II's atmospheric correction process iterative separate the aerosol reflectance and turbid water's reflectance in NIR bands by using spectral relationship of IOPs at 620, 709, 745, 865 nm (Ahn and Park, 2020). The method often fails over highly turbid water because the initial aerosol reflectance spectral slope in two NIR bands exceeds ranges of candidate aerosol models (left image of figure 3). In this update, first guess of the iterative process is directly calculated based on the Ruddick et al. (2000) and the presumed aerosol model (M80) to keep acceptable range within the candidate aerosol models in highly turbid waters (figure 3, right).

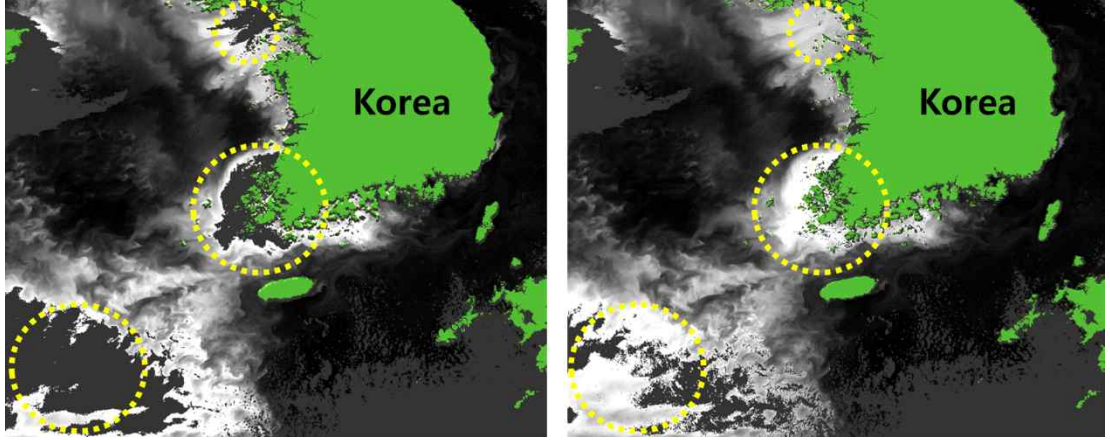


Figure 3. Improvement result of the atmospheric correction over highly turbid waters (dotted circles). The left figure is a result example without the update at 22th/March/2021 03:25 UTC. The right figure is a example of the improvement after the update.

3. Results

The atmospheric correction results after the update are validated by AERONET-OC from the Socheongcho Station located at 37.42313° N, 124.73804° E. The accuracy of R_{rs} has improved by 31.7%, 30.7%, 15.2%, 13.6%, 79.5%, and 50.8% for the 412, 443, 490, 555, 660, and 680 nm bands, respectively (figure 4 and 5).

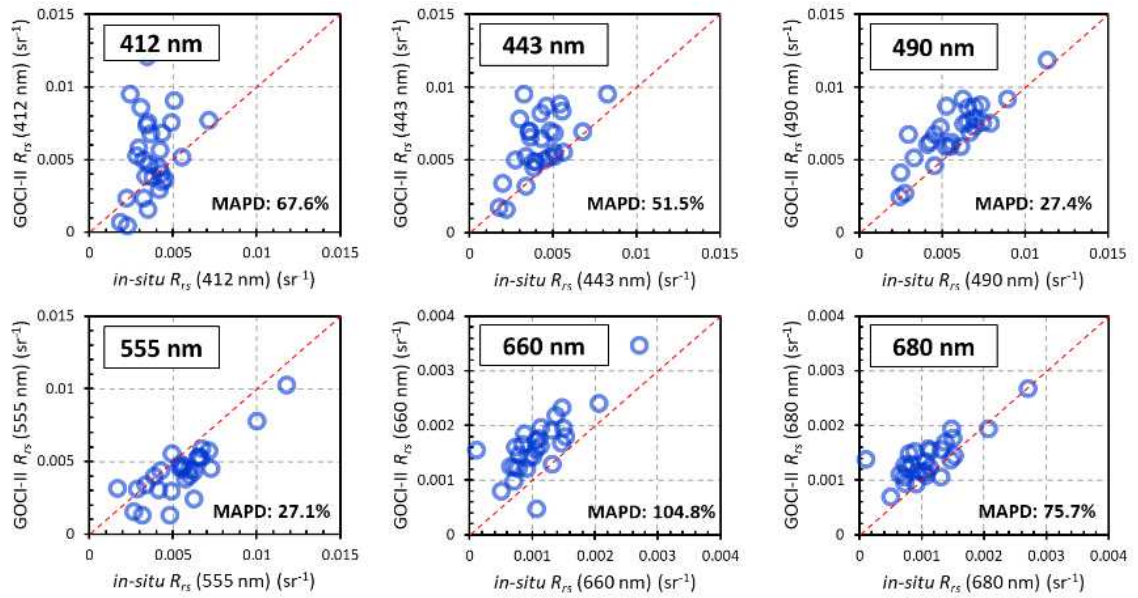


Figure 4. Previous validation results before the update. The GOCI-II R_{rs} is validated by AERONET-OC located at Socheongdo Station.

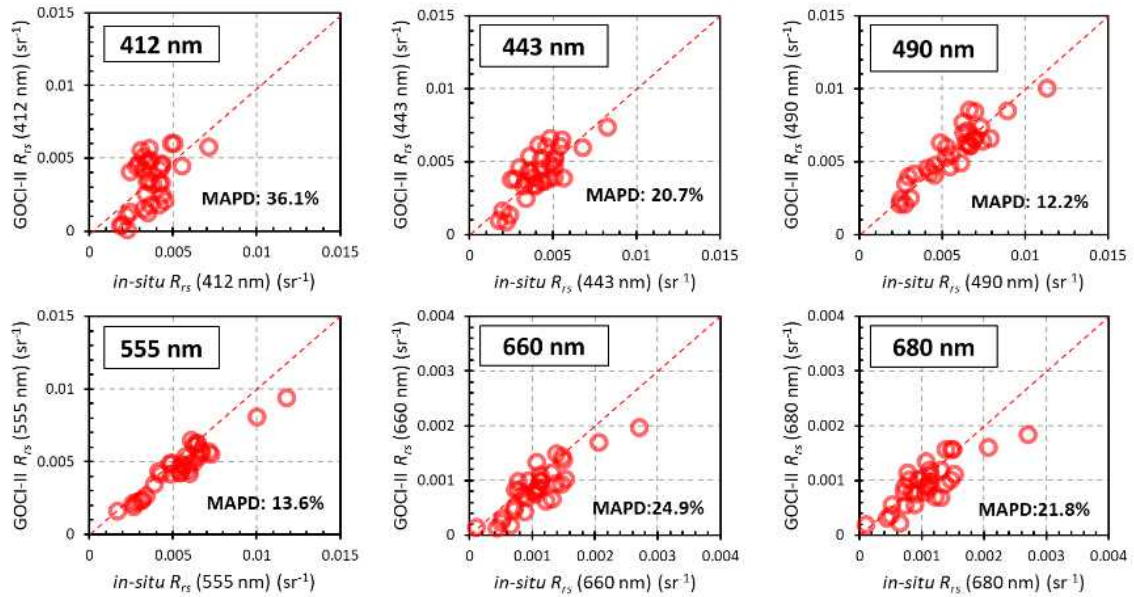


Figure 5. The GOCI-II R_{rs} validation results after the update.

4. Implementation and distribution

The updated products are distributed via webpage (<https://www.nosc.go.kr/>) provided by National Ocean Satellite Center of Korea Hydrographic and Oceanographic Agency after 14/Apr./2024 20UTC (15/Apr./2024 05KST).

[References]

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